MySQL JOINS Practice Database Schema

Use the following schema and sample data before attempting the JOIN questions.

Step 1: Create Database

```
CREATE DATABASE mysql_joins_exam;
USE mysql_joins_exam;
```

Step 2: Create Departments Table

```
CREATE TABLE departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50),
    location VARCHAR(50)
);
```

Insert Data

```
INSERT INTO departments VALUES
(1,'HR','Hyderabad'),
(2,'IT','Bangalore'),
(3,'Finance','Chennai'),
(4,'Marketing','Mumbai'),
(5,'Sales','Delhi');
```

Step 3: Create Employees Table

```
CREATE TABLE employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    gender VARCHAR(10),
    salary DECIMAL(10,2),
    hire_date DATE,
    dept_id INT
);
```

Insert Data

```
INSERT INTO employees VALUES
(101,'Rahul','Male',45000,'2023-01-10',1),
(102,'Priya','Female',60000,'2022-03-15',2),
(103,'Amit','Male',55000,'2021-05-20',2),
(104,'Sneha','Female',50000,'2020-08-11',3),
(105,'Kiran','Male',70000,'2019-09-18',4),
(106,'Anjali','Female',48000,'2022-11-25',NULL),
(107,'Ravi','Male',40000,'2024-01-05',2),
(108,'Pooja','Female',65000,'2021-12-12',5),
(109,'Arjun','Male',52000,'2023-04-08',NULL),
(110,'Deepika','Female',58000,'2022-06-14',3);
```

Step 4: Create Projects Table

```
CREATE TABLE projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100),
    start_date DATE,
    emp_id INT
);
```

Insert Data

```
INSERT INTO projects VALUES
(201,'Hospital Management System','2024-01-10',101),
(202,'E-Commerce Website','2024-02-15',102),
(203,'Payroll System','2024-03-20',104),
(204,'Inventory Management','2024-04-12',105),
(205,'AI Chatbot','2024-05-01',107),
(206,'Banking Portal','2024-06-10',110),
(207,'School ERP','2024-07-05',NULL),
(208,'CRM Application','2024-08-08',108);
```

Step 5: Create Clients Table

```
CREATE TABLE clients (
    client_id INT PRIMARY KEY,
    client_name VARCHAR(100),
    country VARCHAR(50),
    project_id INT
);
```

Insert Data

```
INSERT INTO clients VALUES
(1,'ABC Technologies','India',201),
(2,'Global Soft','USA',202),
(3,'NextGen Solutions','Canada',203),
(4,'Smart Systems','UK',204),
(5,'Tech Innovators','Australia',205),
(6,'Future Corp','Germany',208),
(7,'Digital World','India',NULL);
```

Database Relationship

```
Departments
------------
dept_id (PK)
dept_name
location
     |
     | (1 : Many)
     |
Employees
------------
emp_id (PK)
emp_name
gender
salary
hire_date
dept_id (FK)
     |
     | (1 : Many)
     |
Projects
------------
project_id (PK)
project_name
start_date
emp_id (FK)
     |
     | (1 : Many)
     |
Clients
------------
client_id (PK)
client_name
country
```

project_id (FK)

Tables Summary

Table	Records
Departments	5
Employees	10
Projects	8
Clients	7

MySQL JOINS Practice Question Paper

Total Questions: 30

Duration: 60 Minutes
Level: Fresher / Placement Preparation

Section-A: INNER JOIN (Questions 1-10)

Q1.

Display the employee name and department name.

Q2.

Display employee ID, employee name, and department location.

Q3.

Display all employees working in the **IT** department.

Q4.

Display employee names along with their department names and salaries.

Q5.

Display all female employees with their department names.

Q6.

Display employees who work in the **Finance** department.

Q7.

Display employees whose salary is greater than **55,000** along with department names.

Q8.

Display employee names hired after **2022-01-01** along with department names.

Q9.

Display employee names and department locations in ascending order of employee name.

Q10.

Display employee name, department name, and department location.

Section-B: LEFT JOIN (Questions 11-15)

Q11.

Display all employees along with their department names, including employees without departments.

Q12.

Find employees who are not assigned to any department.

Q13.

Display employee names and department names. If a department is missing, display **'No Department'**.

Q14.

Display all employees and their department locations, including NULL values.

Q15.

Count how many employees are working in each department, including departments with zero employees.

Section-C: RIGHT JOIN (Questions 16-20)

Q16.

Display all departments, including those without employees.

Q17.

Find departments where no employees are currently working.

Q18.

Display department names with employee names using RIGHT JOIN.

Q19.

Display all department names with employee salaries.

Q20.

Display department names in alphabetical order with corresponding employee names.

Section-D: Multiple Table JOIN (Questions 21-25)

Q21.

Display employee names and their project names.

Q22.

Display employee name, department name, and project name.

Q23.

Display project names along with client names.

Q24.

Display employee name, project name, and client name.

Q25.

Display employee name, department name, project name, and client name in a single query.

Section-E: Interview & Advanced Practice (Questions 26-30)

Q26.

Find employees who are not assigned to any project.

Q27.

Find projects that do not have any employee assigned.

Q28.

Find clients who do not have any project assigned.

Q29.

Display the total number of employees in each department. Sort the results by employee count in descending order.

Q30.

Display the following details in a single query:

- Employee ID
- Employee Name
- Department Name
- Project Name
- Client Name
- Salary
- Department Location

Use appropriate JOINs so that employees without departments or projects are also included in the result.